

Fig 2A

$$\boxed{W_M^{1/2}} \rightarrow \boxed{S} \rightarrow \boxed{W_R(c)} \rightarrow \boxed{S^T} \rightarrow \boxed{W_M^{1/2}}$$

Metabolite Laplacian $\Delta_M(c)$

Fig 2C

$D^2 = \text{block-diagonal Laplacians}$

$\Delta_M(c)$	$\Delta_R(c)$
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Fig 2B: Dirac operator

0	$d(c)$
$d^*(c)$ Strong coupling	0

Fig 2D



Weak coupling

